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Facile *ex situ* formation of a LiF–polymer composite layer as an artificial SEI layer on Li metal by simple roll-press processing for carbonate electrolyte-based Li metal batteries†

Seho Sun,[a] Seungcheol Myung,[a] Gaeun Kim,[a] Dongsoo Lee,[a] Hyunsu Son,[a] Minchul Jang,[b] Eunkyung Park,[b] Byoungkuk Son,[b] Yeon-Gil Jung,[c] Ungyu Paik *[a] and Taeseup Song *[a]

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Lithium (Li) metal is considered an ideal anode for next-generation Li metal batteries (LMBs). However, Li dendrite formation and low coulombic efficiency (CE) are major obstacles for its practical use in LMBs. The engineering of the interface between Li metal and electrolytes is crucial for better Li ion kinetics and interface stability to overcome the above problems. In this study, we report a thin LiF/defluorinated polymer (LiF@Po) composite as an artificial SEI layer. The LiF@Po composite layer, formed by a facile and scalable roll-pressing process using polytetrafluoroethylene (PTFE) and Li metal, is composed of an inner (lithium side) LiF-rich layer and an outer (electrolyte side) polymer rich (polyene/fluoropolymer) layer. The free space between LiF particles could be conformally filled with a polymeric layer, improving the Li surface coverage properties of the protection layer. This novel LiF@Po protection layer enables highly stable cyclability even in a carbonate-based electrolyte for more than 1000 h at 1 mA cm^{-2} in a Li||Li symmetric cell and a cycle retention of 80% capacity for >200 cycles with an extremely stable average CE of 99.7% in a Li||$LiCoO_2$ full cell. Furthermore, this LiF@Po protection layer could significantly enhance the cyclability and reversible capacity in a Li||$LiNi_{0.8}Co_{0.1}Mn_{0.1}O_2$ (NCM-811) full cell.

1. Introduction

The demand for Li ion batteries (LIBs) with high energy density has dramatically increased for applications in portable electronic devices and electric vehicles.[1–3] However, the specific energy density of state-of-the-art LIBs employing layered-oxide-based cathode and graphite anode materials has almost reached its theoretical limit.[4–6] Therefore, alternative anode materials with high specific capacity that can replace the graphite anode material have been extensively explored. Among various candidates, lithium (Li) metal has received significant attention as it has the lowest density of all metals (0.53 g cm^{-3}), the lowest negative electrochemical potential (−3.04 V *vs.* SHE), and the highest specific capacity (3860 mA h g^{-1}). Successful employment of Li metal as the anode is also crucial for next-generation rechargeable batteries such as Li–S and Li–O_2 batteries.[7,8] However, the poor electrochemical properties of Li metal and safety issues have hindered the practical utilization of Li metal anodes.[9–11] These problems are mainly attributed to an unstable and inhomogeneous solid electrolyte interphase (SEI) layer formed on the surface of Li metal as a result of the high reactivity of Li with the electrolyte and the large volume change of the Li anode during cycling. Localized high current density and mechanical/chemical defects associated with the inhomogeneous composition and/or cracks in the SEI layer lead to the facile growth of Li dendrites and "dead" Li (electrochemically inactive Li), which increase cell polarization and capacity loss, eventually resulting in cell failure.[12–14] The introduction of a lithium ion-conductive and chemically/electrochemically stable interface layer between the Li metal and electrolyte could significantly improve the electrochemical properties of the cell and its reliability.

Various strategies have been explored to improve the electrochemical properties of Li metal batteries by the formation of a stable SEI layer on lithium metal. The electrochemical properties of the Li metal are significantly governed by the configuration and composition of the SEI layer.[15–20] Lithium fluoride (LiF) has received great attention as a lithium protection layer due to the stable Li plating/stripping behavior resulting from its low Li diffusion barrier energy.[13,19,21–24] Strategies for formation of a LiF-rich film on Li metal can be categorized into *in situ* or *ex*

[a]Department of Energy Engineering, Hanyang University, Seoul, 04763, Republic of Korea. E-mail: upaik@hanyang.ac.kr; tssong@hanyang.ac.kr

[b]Future Technology Research Center, LG Chem., Seoul, 07796, Republic of Korea

[c]School of Materials Science and Engineering, Changwon National University, 51140, Republic of Korea

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situ methods. As an *in situ* method, Markevich *et al.* reported that a fluoroethylene carbonate (FEC)-based electrolyte could improve the electrochemical properties of LMBs due to the formation of a highly crystalline LiF layer on the Li metal surface.[25] It has also been reported that the introduction of a small amount of LiF in the electrolyte could enhance the stability of the lithium metal surface during cycling.[23] Although the *in situ* method is facile and simple, it is difficult to achieve the desirable composition and configuration of the protection layer *via* this method. Unlike *in situ* methods, uniform coating of a Li protective SEI layer on the Li metal surface with reliable controllability of composition and configuration can be realized by using *ex situ* methods. Recently, Cui *et al.* prepared LiF-coated Li metal *via* a direct fluorination process using the fluoropolymer, CYTOP, as a precursor and fluoro-gas and Freon as fluorine sources.[26,27] Also, a facile liquid/solid phase fluorination process using a PVDF-DMF solution has been reported.[28] Although *ex situ* methods show promise, their practical use is limited by the high toxicity of the fluorine gas, requirement of complex equipment, and difficulty in scaling-up for large area processing. Furthermore, although it is important to prepare a protection layer without cracks or pinholes to obtain reliable performance, it is difficult to form a LiF thin film with a pinhole-free structure *via* previously reported methods due to the brittleness of ceramic compounds.[29] Very recently, our group has reported a facile and cost-effective Li metal interface formation process by using roll-pressing between Li metal and copper nitride nanowires which guides Li ion flux and plating/stripping behavior.[30] For successful commercialization of Li metal batteries (LMBs), it is necessary to develop a facile, cost-effective, eco-friendly, and mass and large-area scalable process to synthesize a Li protection layer with reliable performances.

Herein, we report a LiF/defluorinated polymer (LiF@Po) composite as a Li metal protective artificial SEI layer, which is uniformly formed on the Li metal by a simple and scalable roll-pressing of Li metal and polytetrafluoroethylene (PTFE) as a sacrificial layer. The LiF@Po layer is composed of an inner (lithium side) LiF-rich layer and an outer (electrolyte side) polyene and unsaturated fluoropolymer composite polymeric layer. The inner LiF rich layer in contact with Li metal could improve the surface stability of Li metal and Li ion kinetics due to the high mechanical and dielectric strength and the low Li^+ diffusion barrier of the LiF nano-crystalline domain, respectively. The outer polymer rich layer conformally placed on the LiF rich layer could prevent the contact between the Li metal and the electrolyte directly. LMBs employing the LiF@Po-coated Li metal anode (LiF@Po–Li) exhibited a remarkable improvement in electrochemical properties. The symmetric cell shows stable plating/stripping behavior (>1000 hours @ 1 mA cm^{-2}, 1 mA h cm^{-2}), low overpotential (<50 mV @ 500 hours) and low interfacial resistance (<40 Ω of R_{ct}). The LiF@Po–Li||$LiCoO_2$ full cell displayed low voltage polarization (50 mV lower than that of the bare sample @ the first charge) and stable cycle performance (more than 200 cycles at 80% capacity retention). The electrochemical properties of the LiF@Po–Li||NCM-811 full cell are also evaluated to understand the function of the LiF@Po protection layer in a high Ni cathode material from which metal ions dissolved into the electrolyte easily. The cell with the LiF@Po protection layer shows much reduced polarization in the voltage profiles and improved cycle performance compared to the cell without the protection layer. These improvements are attributed to the LiF@Po layer. Moreover, the process is facile, scalable and cost-effective for practical application in Li metal batteries.

2. Results and discussion

2.1. Characterization of LiF domains and carbon structures

Jansta *et al.*[31–33] reported the decomposition of PTFE when in contact with Li-amalgam at ambient temperature based on an electrochemical reaction, in which the Li-amalgam and PTFE act as an anode and cathodic depolarizer, respectively. Since then, several researchers have been attempted to make defluorinated carbon structures using this kind of method which is de-fluorination of PTFE by reaction with alkali metals or their alloys.[34–38] These electrochemical reactions generate metal fluorides and defluorinated polymers such as polyene or/and unsaturated fluorocarbon species. An electrochemical reaction between the Li metal and PTFE leads to the formation of a thin passivation layer composed of LiF and de-fluorinated carbon chains (LiF@Po) like in previous research. Fig. 1 describes the formation process of the LiF@Po layer on the Li metal surface by roll-pressing of PTFE and Li metal. As a result of pressing, the LiF and defluorinated polymers are formed on the interface between the Li metal and PTFE (Fig. 1a). Under these harsh pressing conditions, the mechanical energy generated at the interface between Li metal and the PTFE film plays a role beyond the activation required for the reaction between Li metal and the PTFE film.[33] This mechanical energy-induced fluorination of Li and defluorination of PTFE are shown in Fig. 1b. During the reaction, highly electronegative F atoms react with Li^+, breaking C–F bonds. The appearance of the Li metal foil surface changed from shiny to white and opaque with some fissures which are formed on the Li surface after the roll-press process due to the difference in tensile strength between Li metal and PTFE (Fig. 1c and d; to eliminate macro-structural effects during the overall electrochemical analysis, smooth and flat parts except for the fissures were evaluated). The sacrificial PTFE layer changed from white to deep black after reaction with Li metal due to the defluorination (carbonation) of the PTFE film (Fig. 1e). Fig. 1f shows the color change of the Li metal surface as a function of the reaction time between Li metal and PTFE. When increasing the reaction time, the Li surface gradually darkened, which indicates the formation of polyenes and/or unsaturated fluoropolymer on Li metal. The overall physical optic changes of PTFE and the Li surface after the reaction process imply that the electronic structure of carbon atoms changed from sp^3 to sp^2 or sp^1 with lowered π–π* band gap resulting in better light adsorption properties than of the sp^3 electronic structure and it is matched well with the Raman spectrum of the LiF@Po film layer (Fig. S1†).[39] At the initial stage of the reaction between Li metal and PTFE, LiF nano particles are generated on the Li metal surface. During the

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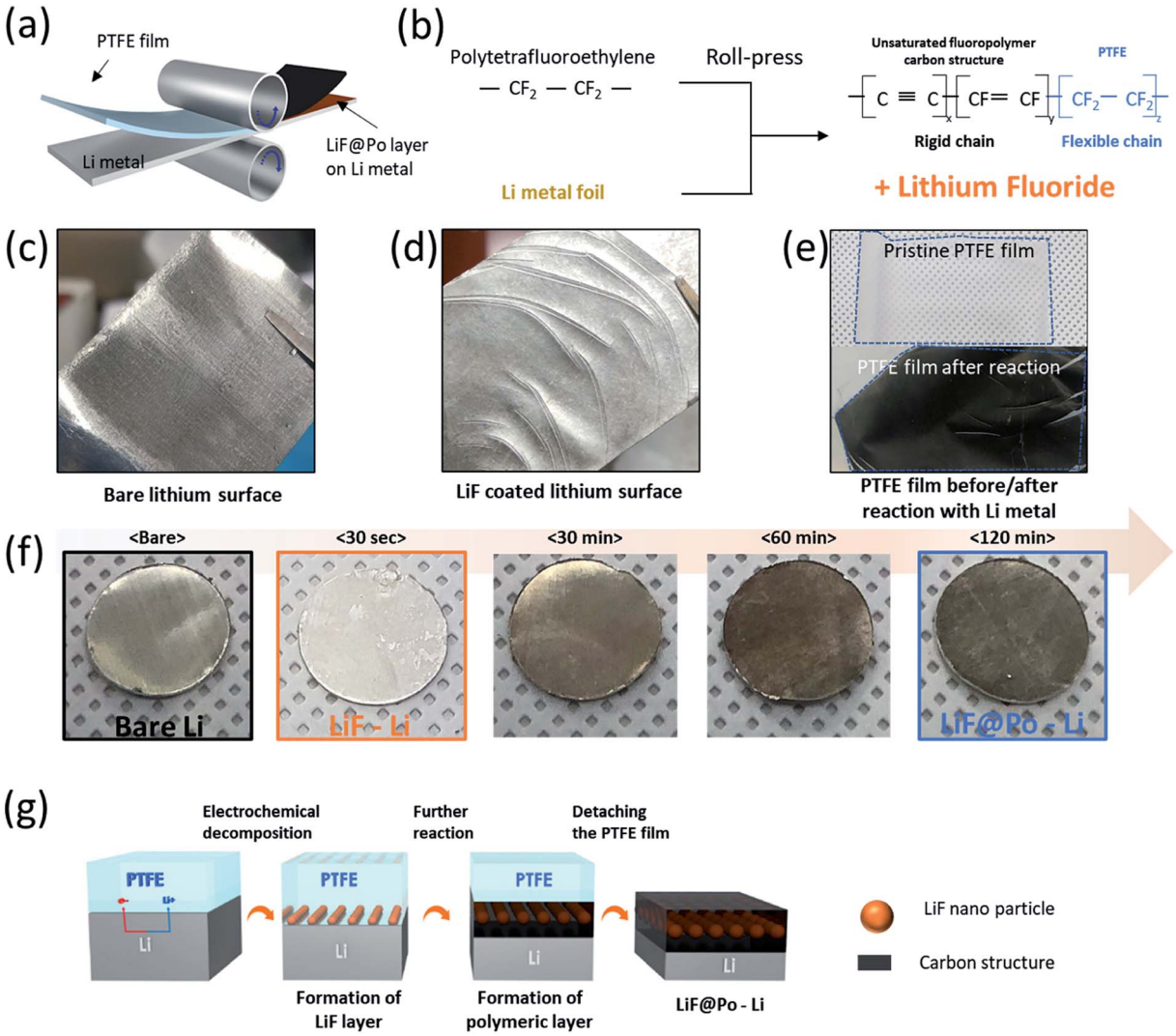

Fig. 1 (a) Schematic illustration of the LiF or LiF@Po protection layer formation on the Li metal by the roll pressing process between Li metal and the PTFE film. (b) Chemical reaction model during the roll pressing process. (c–e) Digital photographs of (c) the bare Li metal, (d) LiF coated Li metal after 30 s of reaction between Li metal and the PTFE film and (e) PTFE film before reaction (upper panel) and after reaction (lower panel) with Li metal. (f) Digital photographs of the Li metal as a function of the reaction time between Li metal and the PTFE film. (g) Schematic illustration of the LiF and polymeric layer formation during the reaction.

reaction, the portion of defluorinated polymers consisting of sp^2 and sp^1 π bonds on the PTFE film surface increases. These defluorinated polymers are more rigid than neat PTFE polymer chains due to their poor C–C bond rotation freedom. Therefore, as the reaction proceeds, more polymeric layers might be easily detached from the PTFE surface (Fig. S1 and S2†).[40,41] Fig. 1g displays the schematic illustration of the formation process of LiF nanoparticles and polymer complexes on the Li metal.

The morphology and the chemical composition change of the Li metal surface as a function of the reaction time (30 s and 120 min) between the Li metal and PTFE film were characterized by scanning electron microscopy (SEM), X-ray photoelectron spectroscopy (XPS), transmission electron microscopy (TEM), and energy dispersive spectrometry (EDS) mapping. As shown in the SEM image of Fig. 2a, the pristine Li metal has a smooth and flat surface. From XPS analysis, the peak intensity related to F1s on the pristine Li is also negligible (Fig. 2d, the peak intensity of pristine Li is magnified to five-times its original intensity). On the other hand, after the reaction for only 30 s between the Li metal and the PTFE film, the surface morphology of the Li metal transformed from flat to rugged (Fig. 2b). The Li metal surface is presumed to be partially covered by the nano-sized LiF, defluorinated PTFE and detached neat PTFE film. The XPS characterization results show existence of Li–F and various C–F bonds (Fig. 2d and e). High peak intensities of the F1s peak at 685.5 eV and Li 1 s peak at 54.5 eV are observed due to the formation of LiF (Fig. 2d and e). As shown in Fig. 2d and f, the peaks related to C–F_2 and PTFE are newly generated after the reaction of LiF–Li. The PTFE peaks in the F1s (~689.5 eV) and C1s (~292.5 eV) XPS spectra of LiF–Li are for detached neat PTFE particles from the PTFE film after roll-pressing due to the insufficient reaction time (from here

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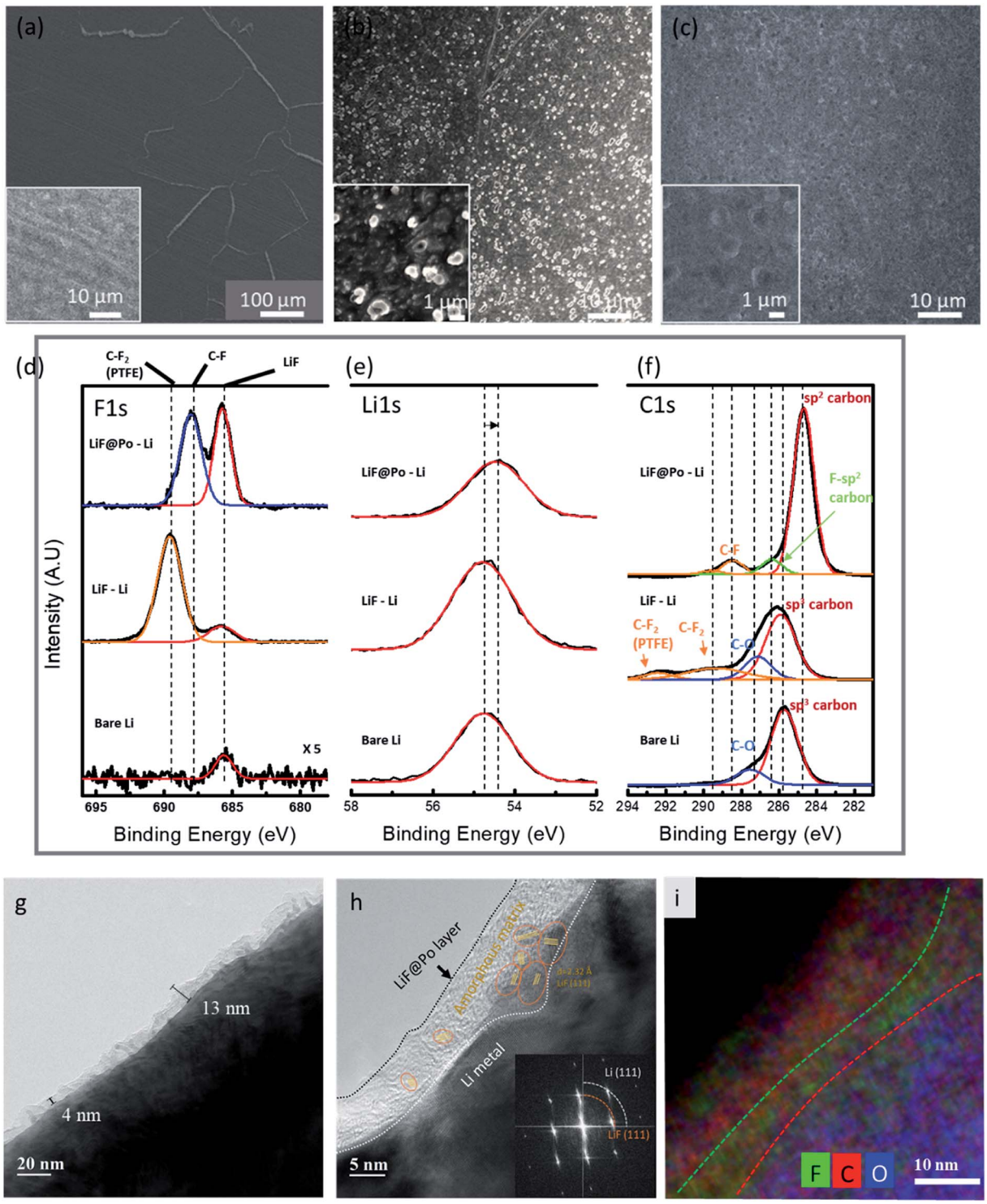

Fig. 2 Scanning electron microscopy (SEM), X-ray photoelectron spectroscopy (XPS) and transmission electron microscopy (TEM) characterization results of the protection layer. (a–c) Top view SEM images of (a) the bare Li, (b) the LiF–Li and (c) the LiF@Po–Li. Inset images show the high magnification SEM images. (d–f) XPS characterization results of F1s (d), Li1s (e) and C1s (f) for bare Li, LiF–Li and LiF@Po–Li. (g) Low-magnification TEM image of the LiF@Po layer on Li metal. (h) High-resolution (HR) TEM image of the LiF@Po–Li. The protection layer is composed of crystalline LiF nano-particles and an amorphous polymer layer. The inset shows the (111) lattice plane of LiF in the fast Fourier transform (FFT) diffraction pattern. (i) Energy dispersive spectrometry (EDS) mapping image of the LiF@Po layer on Li metal. Red, green, and blue dots represent C, F and O elements, respectively. The dominant red color in the outer layer (above the green dotted line) and dominant green color in the inner layer (between the green dotted line and red dotted line) correspond to the polymer rich layer and LiF rich layer, respectively.

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referred to as "LiF–Li" for the 30 s reaction Li sample, where the small portion of polymers is omitted for clarity). After further reaction for 120 min between Li metal and PTFE, the morphology and chemical composition of the Li metal are significantly changed (Fig. 2c). The Li metal surface becomes smoother than that of the LiF–Li, indicating the formation of a denser and more conformal layer composed of LiF and a polymeric layer (from here referred to as "LiF@Po–Li"; LiF domains and a polymer carbon chain layer on Li metal). The LiF@Po–Li shows strong peaks in the F1s spectrum at 688 eV and C1s spectrum at 286 and 288 eV corresponding to the C–F bond compared to LiF–Li, indicating the formation of unsaturated fluoropolymers derived from the PTFE film due to the detaching of defluorinated polymers from the PTFE film (Fig. 2f). Moreover, the peak shift to a lower binding energy of the Li1s peak from 54.8 eV to 54.4 eV is attributed to the existence of LiF and Li metal within the polymeric matrix (Fig. 2e).[42] This is in good agreement with the color change of Li metal as a function of the reaction time. The formation of unsaturated fluoropolymers on the Li metal surface induces a color change (Fig. 1f). As described in Fig. 1g, LiF nano-crystals are formed on the Li metal first, followed by the formation of the polymeric layer composed of rigid polymeric chains such as polyene and unsaturated fluoropolymers, resulting in a uniform Li passivation layer. TEM was employed for further analysis of the LiF@Po–Li layer on Li metal. For sample preparation, Li metal was deposited on the TEM grid by electrochemical deposition.[43] Before the LiF formation process, the Li coated grid was washed with dimethyl carbonate (DMC) solvent to remove the SEI and Li salts on the Li surface (Fig. S3†). Fig. S4† shows that Li is well deposited on the Cu grid. To form the LiF@Po layer on the Li coated Cu grid, the Li coated Cu grid and PTFE film were also roll-pressed for 120 min. The low-magnification TEM image (Fig. 2g) clearly shows the uniform and conformal formation of the LiF@Po layer on the Li metal. The deviation in the thickness might be attributed to the inhomogeneous mechanical force applied to the Li metal and PTFE. High-resolution (HR) TEM images showed ultrafine LiF crystallites with an average size of about 5 nm (Fig. 2h orange circles, Fig. S5†) in the amorphous region which indicate the defluorinated polymeric layers. A lattice with an interplanar spacing of 0.232 nm corresponds to the LiF (111) lattice plane (the inset shows the FFT (Fast Fourier Transform) image in Fig. 2h and S5b†).[27] The grazing incidence XRD (GI-XRD) pattern of the LiF@Po layer also shows the formation of crystalline LiF in the LiF@Po layer (Fig. S5c†). Energy dispersive spectrometry (EDS) mapping was also conducted for elemental analysis. As shown in Fig. 2i, the LiF@Po layer with a 20 nm or less thickness, which is obviously thicker than the SEI layer formed on Li (Fig. S4b†), is composed of fluorine and carbon atoms (red, green, and blue represent C, F, and O, respectively). The LiF@Po composite layer is composed of an inner (lithium side) 5 nm-thick LiF-rich layer and an outer (electrolyte side) polymeric (polyene/unsaturated fluoropolymer) layer, the amorphous region. The observed oxygen atoms originated from the oxidized Li, indicating that Li metal is uniformly covered with a LiF@Po protection layer (Fig. S6†).

2.2. Li plating and stripping behaviors

The plating and stripping behaviors of the LiF@Po–Li electrodes are observed by SEM after the Li deposition of 1 mA h cm^{-2} at a current density of 0.5 mA cm^{-2} with an EC/DEC:3/7, 1.3 M $LiPF_6$, and 5 wt% FEC electrolyte (Fig. 3). All the electrochemical experiments are performed with carbonate-based electrolytes. It has been well known that the ether-based electrolytes, especially the DME/DOL system, exhibit better electrochemical properties compared to carbonate-based electrolytes in LMBs thanks to their stable SEI layer formation and more freedom to choose additives. However, the ether-based electrolyte is not suitable for various commercial cathode

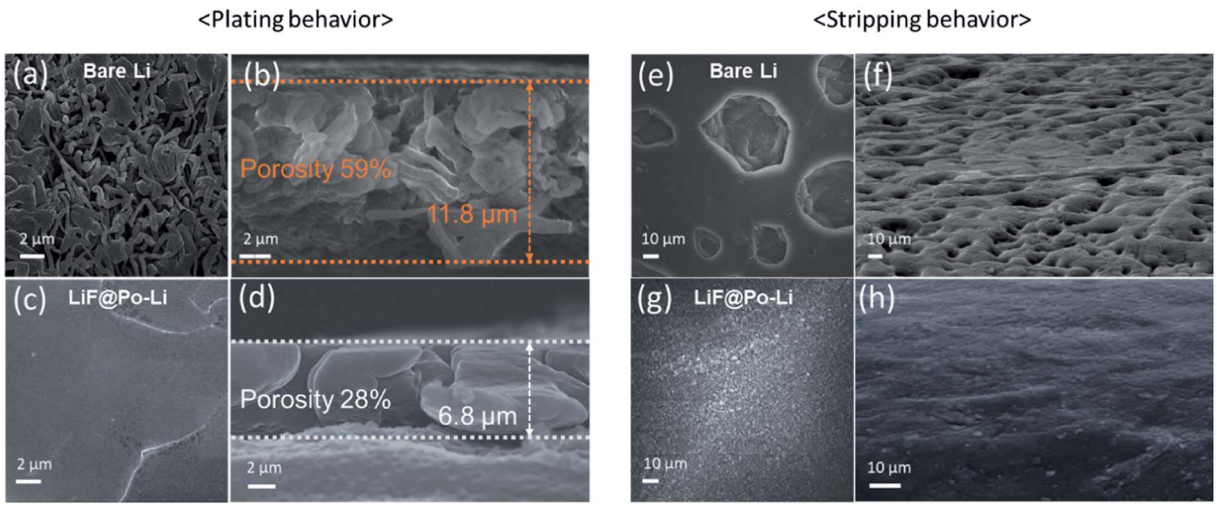

Fig. 3 Li plating and stripping behavior of the bare Li and LiF@Po–Li electrodes. (a and c) Top view SEM images of (a) the bare Li and (c) LiF@Po–Li electrodes after 1 mA h cm^{-2} Li plating, respectively. (b and d) Cross-sectional SEM images of plated Li on (b) the bare Li and (d) LiF@Po–Li. (e and g) Top view SEM image of (e) the bare Li and (g) LiF@Po–Li electrodes after 1 mA h cm^{-2} Li stripping. (f and h) Tilted view SEM images of (f) the bare Li and (h) the LiF@Po–Li electrodes after 1 mA h cm^{-2} Li stripping.

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material due to its instability above 4.0 V *vs.* Li/Li^+.[6] In this regard, carbonated-based electrolytes should be explored for their electrochemical properties in upcoming high energy density LMBs. The electrodes are washed with dimethyl carbonate (DMC) to remove residual salts before SEM observation. The Li metal plated on the bare Li electrode shows a mossy, porous and dendritic morphology with an 11.8 μm thickness and 59% porosity (Fig. 3a, b and ESI Note 1†). In contrast, the Li metal plated on the LiF@Po–Li electrode exhibits a smooth and film-like morphology with significantly reduced porosity (Fig. 3c, d and S7†). The cross-sectional view of the LiF@Po–Li electrode after plating clearly shows that a 6.8 μm-thick dense Li film is deposited on the LiF@Po–Li electrode with 28% porosity (Fig. 3d). The stripping behavior is also monitored. As shown in Fig. 3e and f, the bare Li electrode has high-density craters with sizes ranging from 10 μm to 50 μm after the Li stripping. This non-uniform surface morphology is attributed to the localized current density caused by the inhomogeneous SEI layer. In contrast, the LiF@Po–Li electrode has a smooth and conformal surface without noticeable holes over a large area after Li stripping (Fig. 3g and h). This improvement in plating and stripping behaviors of Li is attributed to the LiF@Po protective layer, which enables a uniform Li ion concentration and current density over the Li electrode by suppressing the electrolyte decomposition.

2.3. Long-term stability and interfacial resistance

To study the effect of the LiF and LiF@Po layers on the electrochemical properties, the long-term cycling properties of symmetric cells with bare Li, LiF–Li and LiF@Po–Li electrodes are evaluated. Fig. 4a shows the voltage profiles for plating/stripping behaviors at a current density of 1 mA cm^{-2} with a plating capacity of 1 mA h cm^{-2} in a carbonate electrolyte (EC/DEC : 3/7, 1.3 M $LiPF_6$, and 5 wt% FEC). The cell with the bare Li electrode (black line) exhibits stable plating/stripping behavior for only 100 hours due to the dendrite growth, polarization caused by dead Li and depletion of the electrolyte. All three electrodes show similar overpotential behavior at the initial stage (Fig. 4b, ~10 cycles). On the other hand, the cell with the LiF–Li electrode (red line) and LiF@Po–Li (blue line) electrode shows a much improved stability compared to the cell with the bare Li electrode. However, significant increase in the polarization is observed in the cell with the LiF–Li electrode after 150 cycles (Fig. 4d). Although LiF on the Li metal surface enables lowering the Li diffusion barrier energy and suppressing dendrite formation, the parasitic reaction between the Li metal and the electrolyte, the deficient passivation coverage and poor flexibility of the LiF layer result in inhomogeneous SEI formation and electrolyte depletion (Fig. 2b).[13,23,27,44] Moreover, it should be noted that extremely stable voltage profiles over 1000 hours without significant increase in the polarization could be achieved in the cell with the LiF@Po–Li electrode. The excellent performance of the LiF@Po–Li electrode is attributed to the low Li diffusion barrier energy originating from the LiF domains and also the conformal coverage of the polymeric layer without pinholes, which enables highly reversible Li plating/stripping while maintaining a dense structure and the formation of the homogeneous SEI layer. Fig. 4e and f show the Nyquist plots for the bare Li, LiF–Li and LiF@Po–Li electrodes before and after cycling. Before cycling, the cell with the LiF–Li electrode shows the highest interfacial resistance (charge transfer resistance, R_{ct}) among all the cells, which is presumed to be due to the presence of the detached neat PTFE. The LiF@Po–Li electrode shows the lowest interfacial resistance before cycling. The uniform nano-crystalline LiF particle and polyene/unsaturated fluoropolymer complex could alleviate the barrier of the Li ion conduction path. It has been reported that a thin polymeric interface layer between the Li metal and the electrolyte could promote Li-ion kinetics by lowering the activation energy for the Li-ion de-solvation.[45–47] After 1 cycle, the LiF–Li electrode exhibits similar interfacial resistance compared to the LiF@Po–Li electrode due to the formation of the SEI layer on the surface of LIF domain decorated Li metal. Both the LiF–Li and LiF@Po–Li electrodes show much lower R_{ct} after cycling compared to the bare Li electrode, which is attributed to improved Li ion kinetics resulting from LiF domains and uniform SEI layer formation. To monitor the stability and electrolyte-proof properties, time-dependent EIS measurements were carried out (Fig. S8†). The bare Li electrode shows a significant increase in charge transfer resistance over time. In the case of the LiF–Li electrode, the charge transfer resistance slightly increased after 24 hours. On the other hand, the LiF@Po –Li electrode exhibited a gradual decrease in the charge transfer resistance with time, which confirms that the LiF@Po layer plays an important role in the stabilization of the interface between Li metal and the electrolyte.

2.4. Electrochemical properties of the full cells

To understand the effect of the Li protection layer on the electrochemical properties of the full cell, the LiF@Po–Li and bare Li anodes were paired with $LiCoO_2$ (LCO) and nickel-rich $LiNi_{0.8}Co_{0.1}Mn_{0.1}O_2$ (NCM-811) cathodes. 500 μm-thick lithium was employed as the anode for the fabrication of the full cell. The N/P ratios of full cells employing LCO and NCM cathodes are 30 and 40, respectively. The LCO shows robust cycling performance due to its stable intercalation mechanism. Therefore, it is possible to study the role of the Li protection layer on the electrochemical properties by minimizing the influence of the cathode material. The NMC-811 exhibits a high reversible capacity of ~190 mA h g^{-1}. However, the structural instability of the high nickel NCM cathode material results in the dissolution of the transition metal from NCM into the electrolyte. Therefore, by coupling with the NCM-811 cathode, the stability of the LiF@Po–Li electrode could be evaluated in the circumstances where the transition metal is eluted in the liquid electrolyte. The cycling performances of the Li||LCO and LiF@Po–Li||LCO full cells were evaluated at a current rate of 1C charge/1C discharge of the LCO cathode and 0.5C charge/1C discharge of the NCM-811 cathode, corresponding to a high current density of 3.4 mA cm^{-2} (21 mg cm^{-2} of the LCO cathode and 3.4 mA cm^{-2} plating/stripping of the Li anode) and 3 mA cm^{-2} (13.7 mg cm^{-2} of the NCM-811 cathode and 1.5 mA cm^{-2}

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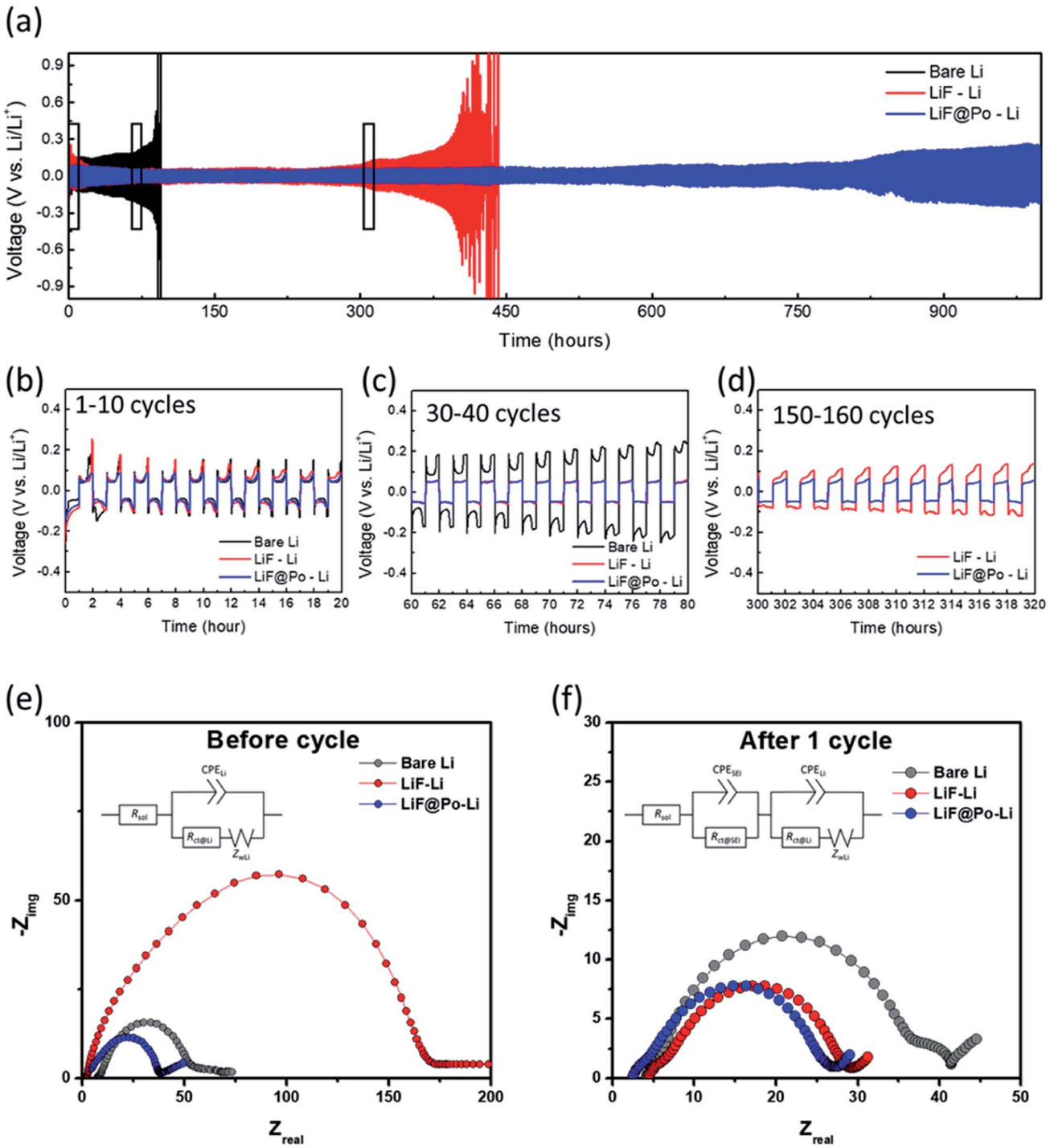

Fig. 4 Electrochemical properties of Li symmetric cells with the bare Li, LiF–Li and LiF@Po–Li electrodes. (a–d) Galvanostatic charge–discharge voltage profiles of Li symmetric cells with the bare Li (black), LiF–Li (red), and LiF@Po–Li (blue) electrodes with a 1.3 M $LiPF_6$ in EC/DEC : 3/7 with 5 wt% FEC electrolyte. The current pulse was ±1 mA cm^{-2} for each 1 h cycle. (e and f) Nyquist plots of the cells (e) before and (f) after 1 cycle. The inset images show the equivalent circuit of Li symmetric cells before and after 1 cycle, respectively.

plating and 3 mA cm^{-2} stripping of the Li anode), respectively (Fig. 5). The Li‖LCO and LiF@Po–Li‖LCO full cells deliver reversible capacities of 146.7 mA h g^{-1} and 153.3 mA h g^{-1}, respectively. The Li‖LCO full cell exhibits a dramatic capacity fading after 177 cycles, which indicates Li metal depletion corresponding to the sharp C.E drop of the cathode (Fig. 5b). The average coulombic efficiency of the bare Li electrode is 81.6%, which is in good agreement with previous research (ESI Note 2 and eqn (4-1)).[13,48] On the other hand, the LiF@Po–Li‖LCO full cell runs for 300 cycles without Li depletion despite the gradual capacity fading of the LCO cathode.[22] The LiF@Po–Li‖LCO cell shows lower voltage polarization than the Li‖LCO

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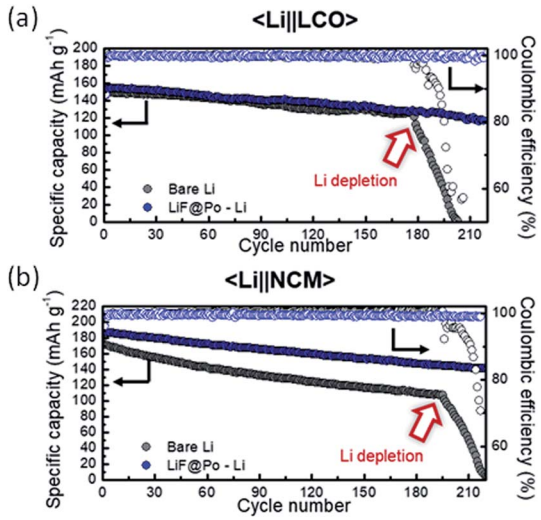

Fig. 5 Electrochemical properties of the full cells with the LCO or NCM cathodes and the bare Li or LiF@Po–Li anodes. (a) Cycle performance of the cells with the LCO cathode (3.4 mA h cm^{-2}) and the bare Li (black circles) or LiF@Po–Li (blue circles) anodes with a 1.3 M $LiPF_6$ in EC/DEC : 3/7 with 5 wt% FEC electrolyte. The current density for charge and discharge was 3.4 mA cm^{-2} (1C). (b) Cycle performance of the cells with the NCM-811 cathode (2.5 mA h cm^{-2}) and the bare Li (black circles) or LiF@Po–Li (blue circles) anodes with a 1 M $LiPF_6$ in EC/DEC/DMC : 1/2/1 with 10 wt% FEC electrolyte. The current densities for charge and discharge were 1.25 mA cm^{-2} (0.5C) and 2.5 mA cm^{-2} (1C), respectively.

cell during the first charge process, which is well corelated with the result in Fig. 4e (Fig. S9†).[23] The rate capabilities were also evaluated at current densities of 0.5C, 1C, 2C and 3C (Fig. S11†). The LiF@Po–Li||LCO cell showed much improved rate capability compared to Li||LCO at 2C and 3C (6 mA cm^{-2} and 9 mA cm^{-2}). It should be noted that the Li||LCO cell shows fast capacity degradation when the current density is changed from 3C to 0.5C while the LiF@Po–Li||LCO cell shows stable capacity retention. The Li||NCM-811 and LiF@Po–Li||NCM-811 full cells show reversible capacities of 172 mA h g^{-1} and 187 mA h g^{-1}, respectively. The Li||NCM-811 full cell shows fast capacity and coulombic efficiency fading after 198 cycles due to Li metal depletion. The average coulombic efficiency of Li metal in the Li||NCM-811 cell is calculated to be 75.2% (ESI Note 2 and eqn (4-2)†) which is lower than that of the LCO full cell resulting from transition metal dissolution.[49] The LiF@Po–Li||NCM-811 cell shows much improved cycle stability with a higher specific capacity than that of the Li||NCM-811 cell for 300 cycles. The typical galvanostatic discharge/charge voltage profiles of the Li||NCM-811 full cell (Fig. S11†) also show relatively higher voltage polarization during the first charge process than those of the cell with LiF@Po–Li. It should be noted that the Li||NCM-811 cell shows poor cycle stability compared to the Li||LCO cell, resulting from the transition metal ions in the liquid electrolyte from the NCM cathode (average coulombic efficiencies of bare Li are 81.6% when paired with LCO *vs.* 75.2% with NCM-811, ESI Note 2, eqn (4)†). Even with the polluted electrolyte (transition metal dissolved in the electrolyte), the LiF@Po–Li||NCM-811 cell exhibits stable cycle performance due to the superior protective performance of the LiF@Po protection layer. To clearly observe the effect of the LiF@Po protection layer, the electrochemical properties of Li metal full cells using thin bare Li metal or LiF@Po–Li electrodes with a thickness of 50 μm and the NCM cathode with a high energy density of 7.2 mA h cm^{-2} (N/P ratio of 1.45) were evaluated (Fig. S12†). Bare Li||NCM and LiF@Po–Li||NCM full cells showed an efficiency of 87% and LiF@Po–Li (50 μm)||NCM (7.2 mA h cm^{-2}) showed average coulombic efficiencies of 87% and 95%, which indicates that the LiF@Po protection layer enables the formation of a stable interface between the Li metal and the electrolyte.

3. Conclusions

In summary, we developed a formation method of a Li protective artificial SEI layer to form a LiF and polymeric composite protection layer directly on lithium metal foil using the electrochemical reaction between the Li and the sacrificial PTFE film by using roll-pressing. The roll-press process is facile, cost-effective, eco-friendly and a mass and large-area scalable process. The composite protection layer composed of a LiF-rich layer and polyene/unsaturated fluoropolymer rich layer could significantly improve the electrochemical properties of LMBs. The LiF@Po protection layer provides a stable interface between the Li metal and the electrolyte, enabling improved Li plating/stripping behavior and ultra-long cycling stability in both the Li||Li symmetric cell and Li||LCO/Li||NCM full cells. In particular, the full cell with the NCM-811 cathode and LiF@Po–Li anode exhibits reasonable electrochemical performance even with transition metal dissolution. The material design strategies presented here could provide important avenues for forming an artificial SEI layer in various applications not only in Li metal systems but also in other metal systems including sodium and potassium.

Conflicts of interest

There are no conflicts to declare.

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